

Field estimation of specific yield in a central Iowa crop field†

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Abstract:

The specific yield (Sy) is the gain or loss of water associated with a corresponding amount of water table rise or fall. Rain may be intercepted by the canopy, refill the soil pore system, taken up by plant roots or drained. Transpired soil water may be replenished from a shallow water table, so water table loss is indirectly tied to evapotranspiration (ET). The purpose of this study was to use continuous data for the different components of the water cycle to back-calculate Sy for both wetting and drying conditions. Two time periods were considered for a toeslope site in an Iowa corn (*Zea mays* L.) field: growing season from 22 June to 10 July 2007, and post-growing season 13–30 October 2007. The field was instrumented with eddy covariance instrumentation for ET estimates, tipping bucket raingage for precipitation, automated well-depth recorder for water table depth and water content reflectometers (CS616s) for soil water content. During drying in the growing season, mean Sy was 0.19, but drying Sy could not be determined post-growing season because of no net upward water movement at this time. During wetting, estimates of Sy were improved when soil water increase was subtracted from total rainfall; then the mean wetting Sy was 0.052 versus 0.31–0.46 without this correction. Large differences between seasonal and post-growing season data reflected the seasonal strong upward gradient created by soil dried from water uptake. Post-growing season data was dominated by drainage, but seasonal root water uptake intercepted drainage water. Published in 2010 by John Wiley & Sons, Ltd.

KEY WORDS specific yield; water flux; evapotranspiration; infiltration

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INTRODUCTION

Specific yield (Sy), defined as the volume of water released under gravity from storage per unit cross sectional area per unit decline in water table (Freeze and Cherry, 1979), is a critical component in many studies of hydrological processes occurring at a shallow water table. For example, water table fluctuations are commonly used to estimate groundwater recharge and the method relies on Sy to convert water table rise to recharge (R) (Healy and Cook, 2002; Heppner *et al.*, 2007):

$$R = Sy \Delta h / \Delta t \quad (1)$$

where, h is water table height and t is time. Similarly, water loss from the water table is also estimated using Sy to convert diurnal water table fluctuations to daily evapotranspiration (ET) (Engel *et al.*, 2005; Loheide *et al.*, 2005; Butler *et al.*, 2007; Schilling, 2007). However, despite its prominent role in estimating major hydrological processes, estimates of Sy are prone to uncertainties (Healy and Cook, 2002).

Specific yield is generally defined in relation to specific retention (Sr) via total porosity (TP) (Healy and Cook,

2002):

$$Sy = TP - Sr \quad (2)$$

In theory, Sr should be an instantaneous release of water, but in practice soil water is released slowly. The concept is considered similar to field capacity (FC), but both require a deep water table and are invalidated by upward water movement from the water table (Cassel and Nielsen, 1986; Healy and Cook, 2002; Chen and Hu, 2004; Shah *et al.*, 2007). Rather than defining Sy in relation to Sr, it would be more accurate to define Sy as satiated water content minus FC, with satiated water content accounting for trapped air at depths below the zero-matric potential level. More specifically, Sy is the gain or loss of water associated with a corresponding amount of water table rise or fall. The difficulty in determining Sy is in quantifying how much actual water is gained or lost, which is complicated by trapped air, non-equilibrium and non-steady-state conditions, hysteresis, etc.

The complexity of the groundwater–soil–crop–atmosphere system is often grossly simplified in estimations of Sy. Water table depth is simultaneously affected by lateral flow in and out, preferential drainage during a rainfall event, matrix draining after a rainfall event, plant water use directly from the capillary fringe of the water table, delayed water movement into the root zone above the water table to replenish plant root uptake, tile drainage and hysteresis differences for rise and fall of

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the water table (Healy and Cook, 2002; Loheide *et al.*, 2005). Loheide *et al.* (2005) provided various calculation procedures for different conditions and suggested that S_y could be estimated from soil texture when water table depths are greater than 1 m and sediment texture is more than 10% fines. Others have noted that the ratio of water table rise to total rainfall can be used as a proxy measure of S_y (Loheide *et al.*, 2005; Risser *et al.*, 2005; Schilling and Kiniry, 2007). The amount of water table rise measured after a rainfall event provides an estimate of the amount of open pore space available in the unsaturated zone. Healy and Cook (2002) outlined other laboratory and field methods to estimate S_y . Along with various aquifer tests, geophysical methods and a water-balance method, Healy and Cook (2002) suggested that a water-budget method could be used to estimate S_y . They noted that S_y could be estimated from the equation:

$$S_y = (P + Q_{in} - Q_{out} - ET - \Delta S) / \Delta h / \Delta t \quad (3)$$

where, P is precipitation, Q_{in} and Q_{out} are flow in and out of the domain, ET is evapotranspiration, S is change in water storage and $\Delta h / \Delta t$ is the change in head over time. Nachabe (2002, 2004, 2005) noted day–night differences for changes in water table depths and changes in the quantity of water in the soil profile. They used these day–night differences to quantify lateral losses and gains of water.

Plants start to use soil water as soon as atmospheric demand increases after a rain event. Less energy is expended to take up water from larger pores than smaller pores, so roots intercept some of the drainage water (Wilcox, 1962; Miller and Aarstad, 1973), which causes drainage to cease earlier under a crop than under a bare area. So the rain-induced water table rise will usually be higher and more rapid for a bare area because plants are not actively taking up water. Zhang and Schilling (2006) observed the water table under bare ground rising 674 mm in 8 h following 86 mm of rainfall compared with less than 30 mm water table rise during the same period under a dense canopy of cool season grass. Further, if plants are actively taking up water, the soil–water system never reaches equilibrium, even during extended periods between rains.

Many studies conveniently assume a constant value for S_y or assume S_y varies systematically according to conditions described by equilibrium equations. Although a water budget method to estimate S_y is not new, it is rarely done because values for ET and ΔS are difficult to obtain with much certainty. The purpose of this study was to use automated data collection systems to estimate S_y for both wetting and drying conditions during early growing season and post-growing season time periods at a field site in central Iowa. Our objectives were to use measured values of ET and ΔS to evaluate how S_y may vary according to plant water demands, soil-water dynamics and water table depth.

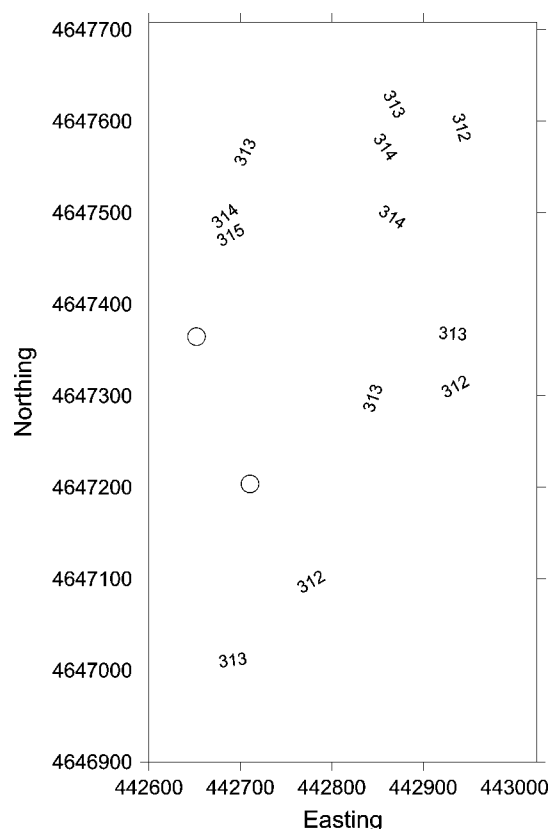


Figure 1. Field of study. The north point is the site for monitoring soil water and water table depth. The south point is the meteorological site

MATERIALS AND METHODS

Field setup

A central Iowa field was in the corn (*Zea mays* L.) year (2007) of a corn–soyabean [*Glycine max* (L.) Merr.] rotation. The corn was planted on 11 May 2007. central Iowa is in the Des Moines lobe, for which landscapes are typically closed-depression topography. The site examined (Figure 1) was at the edge of a toeslope and was a Webster soil (fine-loamy, mixed, superactive, mesic Typic Endoaquolls) (Soil Survey Staff, 2008). The mean sea elevation was 313.5 m, the slope was 3.1%, the depth of the mollic epipedon was 0.79 m, the depth to carbonates was 1.8 m, and a sand lens was evident around 1–1.4 m depth (Table I). Average texture through the profile consisted of 44% sand, 31% silt and 25% clay. East of this site, a tile drained a low area to the southeast; the tile placement passed to within about 50 m south of the site. These approximate tile positions were not field-verified.

Instrument setup

A neutron access tube and monitoring well were installed on 8 June 2007. Neutron probe calibrations are described in Logsdon *et al.* (2009). Neutron probe water contents were read every 1–4 weeks, beginning at 0.3 m depth, and downward every 0.2–2.3 or 2.5 m. Volumetric surface samples were collected to 0.3 m depth, and subdivided every 0.1 m for measurement of volumetric soil water content. The hydrometer method was used for

Table I. Soil properties at the monitored site

Depth	Sand	Silt	Clay	Colour	Other
(m)	(g/g)	(g/g)	(g/g)		
0–0.2	0.377	0.352	0.271	10YR2/1	
0.2–0.4	0.323	0.341	0.336	10YR2/1	
0.4–0.6	0.323	0.343	0.334	10YR2/1	
0.6–0.8	0.310	0.355	0.335	10YR3/1	
0.8–1.0	0.410	0.308	0.282	10YR4/1	
1.0–1.2	0.617	0.192	0.192	10YR5/1	
1.2–1.4	0.711	0.121	0.171	10YR5/2	
1.4–1.6	0.500	0.329	0.171	10YR5/1	Mn
1.6–1.8	0.353	0.447	0.200	10YR5/1 7.5YR4/6	CO ₃

Mn refers to manganese concretions.

CO₃ refers to carbonates.

particle size analysis (Gee and Bauder, 1986) for soil taken from inside the neutron probe access tube during installation. Barometric pressure was collected as part of the micro-meteorology data. Automated well-depth readings were measured with a miniTROLL transducer (non-vented), and readings were corrected for barometric pressure (Rasmussen and Crawford, 1997). Manual water table depths were measured with a well depth reader the same time as the neutron probe soil water contents were measured, and were used for calibrating the well transducer.

CS616 water content reflectometers (Campbell Scientific, Inc., (CSI) Logan, UT) and copper-constantan thermocouples were installed on 22 June 2007 horizontally for 0.05 and 0.15 m depths in both row and inter-row positions. The deeper probes were not installed horizontally because the needed large access pit would remove a large section of the planted crop. Instead a smaller access hole was used to install the probes at an angle between row and inter-row positions, at depths of 0.3, 0.5, 0.7, 0.9 and 1.1 m. The CS616 were offset spatially so a probe was not placed immediately below another probe. The CS616 were read once every hour, and the output was converted to the square root of apparent permittivity ($\epsilon_a^{1/2}$) as described by Kelleners *et al.* (2005). Separate calibrations were needed for wet, dry and cool conditions of each depth (Logsdon *et al.*, 2009). Temperature corrections were needed for 0.05 and 0.15 m depths.

Rainfall was measured with a tipping bucket rain-gage (CSI, Logan, UT) (Figure 1 for meteorological site). Evapotranspiration (latent heat flux) was measured by eddy covariance as described by Hatfield *et al.* (2007). The equipment included a LI-7500 open-path water vapour–carbon dioxide sensor (Li Cor Bioscience, Lincoln, NE) combined with a CSAT sonic anemometer (CSI, Logan, UT). Latent heat flux values were converted to millimeter ET using a conversion factor of $0.381 \mu\text{m W}^{-1} \text{m}^{-2}$ per 15 min interval (given as 15 min totals). The eddy covariance data were screened as described by Hatfield *et al.* (2007), which resulted in some data outside the allowable ranges. Gap-filling of missing ET data was done using an iterative interpolation technique

(Hernandez-Ramirez *et al.*, 2009; Logsdon *et al.*, 2009). Because the rainfall data had a few missing points, the data were combined with rainfall data from an adjacent field to produce a complete record.

Approximately after day 195 in 2007, the water table was deeper than the CS616 probe depths. The water table depth was not within the CS616 probe depth range again until intense rain on days 286–287 and 290–291. These two time periods were examined, i.e. the early crop season (actively transpiring plants, days 173–191 or 22 June to 10 July 2007), and post-crop growing season (no plant transpiration, day 286–303 or 13–30 October 2007). These time periods were chosen because the water table was within or just below the CS616 probe depths.

Specific yield calculations

Table II shows the calculation steps for Sy between rains, which were more extensive than the simple form of Equation (3). Day–night differential has been used to subtract the effect of lateral additions or losses (Healy and Cook, 2002; Nachabe *et al.*, 2004, 2005; Loheide *et al.*, 2005). In our study, we used night-time soil water loss rate (ΔS_n) as an estimate of drainage loss rate (either vertical or lateral) or lateral gain. The daytime hours of ET were consistently high with no negative values, whereas night-time values were small with some negative values. The transitions between day and night varied by day and were determined by inspecting the data, and the same transition times between day and night were used for soil water loss rate.

It was necessary to subtract any measured night-time ET_n rate from ΔS_n :

$$\Delta S_n \text{ net} = \Delta S_n - \text{ET}_n. \quad (4)$$

This correction allowed ΔS_n to be an estimate of drainage rate for both night and day. A negative adjusted ΔS_n actually indicated a net ΔS_n gain (negative drainage due to upward water flux or lateral addition). Note that drainage refers to loss from the unsaturated zone with no implications for how much was merely lost vertically downward or how much eventually moved within the water table or capillary fringe to the tile.

The ΔS_n net was subtracted from the soil water loss rate during the day (ΔS_d) to estimate net transpiration loss excluding drainage loss (Loheide *et al.*, 2005):

$$\Delta S_d \text{ net} = \Delta S_d - \Delta S_n \text{ net}. \quad (5)$$

Negative rates for daytime ΔS_d net were not used further in the calculations, and Sy was not calculated for these days. Negative ΔS_d net indicated that drainage dominated on those days; therefore, a depth of water corresponding to water table loss would not be appropriate.

Next the ΔS_d net rates were subtracted from the ET_d rates:

$$\text{Up} = \text{ET}_d - \Delta S_d \text{ net}. \quad (6)$$

This was similar to the approach of Shah *et al.* (2007) who determined ground water contribution to ET as total

Table II. Calculation of Sy between rains on days 173 and 190

Day	ET _d	ET _n	ΔS _d	ΔS _n	ΔS _n net (ΔS _n – ET _n)	ΔS _d net (ΔS _d – ΔS _n net)	Up (ET _d – ΔS _d net)	ΔWT _{drp}	Sy, Up/WT _{drp}
	(mm d ⁻¹)	(mm d ⁻¹)	(mm d ⁻¹)	(mm d ⁻¹)	(mm d ⁻¹)	(mm d ⁻¹)	(mm d ⁻¹)	(mm d ⁻¹)	
174	4.63	0.37	5.99	6.21	5.84	0.15	4.48	38.28	0.117
175	4.65	0.09	5.08	2.24	2.15	2.93	1.72	25.00	0.066
176	7.73	–0.01	2.24	–0.01	0.0	2.24	5.49	25.07	0.219
177	10.26	0.37	0.80	1.01	0.64	0.16	10.10	21.59	0.468
178	7.19	0.44	6.36	0.99	0.55	5.81	1.38	22.82	0.060
179	5.20	0.21	3.33	4.75	4.54	–1.21	6.41	20.51	0.313
180	5.46	0.03	2.25	4.82	4.79	–2.54	8.00	20.37	0.393
181	8.56	–0.03	3.59	3.04	3.07	0.52	8.04	30.93	0.260
182	9.29	–0.01	5.08	1.54	1.55	3.53	5.76	24.62	0.234
183	9.15	0.58	4.69	0.45	–0.13	4.82	4.33	24.81	0.175
184	9.25	0.62	3.61	–0.42	–1.04	4.65	4.60	25.86	0.178
185	9.72	0.20	6.00	–0.02	–0.22	6.22	3.50	27.13	0.129
186	10.52	0.31	7.29	–0.10	–0.41	7.70	2.82	26.19	0.108
187	9.74	0.05	6.61	–0.26	–0.31	6.92	2.82	20.60	0.137
188	11.93	1.00	8.22	0.57	–0.43	8.65	3.28	33.73	0.097
189	10.61	0.54	5.64	–0.62	–1.16	6.80	3.81	36.00	0.106

ET is evapotranspiration rate, S is soil water loss rate, WT is water table depth. Cut-off between day and night were determined from ET data. The upward rate of water movement is Up. Subscripts refer to day (d), night (n) and drop (drp); net refers to net changes. The Sy is the calculated specific yield.

ET minus the soil water loss in the unsaturated zone; however, they did not include lateral drainage effects as they simulated a level site. An Up value indicated net upward water movement, whereas a negative flux indicated net drainage from the soil profile (0–1.1 m). Finally to determine Sy (Healy and Cook, 2002), the Up values were divided by the 24-h water table drop rate (WT_{drp}) both in mm d⁻¹:

$$Sy = Up/WT_{drp}. \quad (7)$$

During rain periods, Sy was estimated by the amount of water rise in the water table divided by the rainfall depth (RD) (Healy and Cook, 2002; Schilling and Kiniry, 2007). Because some of the rain wets up the soil before causing much water table rise, the amount of soil wetting (ΔS_{wet}) should first be subtracted from the RD. The maximum change in water table from before the rain to after the rain, ΔWT_{ris}, was determined and the total amount of rain was summed during this time. Then the amount of soil water gain was subtracted from the rainfall amount. Adjusted rainfall divided by ΔWT_{ris} provided an estimate of Sy:

$$Sy = (RD - \Delta S_{wet})/\Delta WT_{ris}. \quad (8)$$

RESULTS AND DISCUSSION

Specific yield estimated from ET

During 2 weeks without rain in late June to July (early crop season), there was only a slight day–night difference in the water table loss rate (Figure 2). Day–night differential soil water loss rate became apparent around 9 days after the last rain and was apparent for evaporation in just a day or two after the rain. The water table

might have been too deep for apparent day–night differences caused by lateral flow, or perhaps lateral additions equalled lateral losses. Lateral drainage was less important during the growing season because most soil water was lost in transpiration.

Early after the rain on day 173, the calculated Sy were large because some of the water loss was due to drainage (downward loss of water from the unsaturated zone) (Table II). Daily Sy estimated during the non-rain period was variable, ranging from 0.06 to 0.47. Over time, the calculated Sy values generally decreased. The mean Sy for days 174–189 was 0.19, and for days 181–189 was 0.17 (which eliminated the early drainage loss). At the conclusion of monitoring, the Sy appeared to be approaching an asymptotic value near 0.1.

After the growing season, cumulative ET consisted of evaporation only since plants were not transpiring. During this period, soil water loss was over four times that of evaporation loss (Figure 3). Lateral drainage was likely responsible for much of the soil water and water table loss, although minor diurnal ET variations were still evident. Because ET was much less post-growing season, vertical and lateral drainage were major components of water loss. The sandy layer ~1–1.5 m deep might have facilitated movement to the downslope tile once the water table rose above the tile depth (~1.1 m deep). Diurnal variations in soil water loss and water table fluctuations were not obvious. Estimating Sy during the post-growing season period were problematic because net daytime soil water loss was negative (indicating a net gain in soil water). The calculation procedure was no longer valid because net water movement up from the water table was not detected; therefore, Sy estimated for the post-growing season was based on the Sy determined during rainfall periods (~0.05).

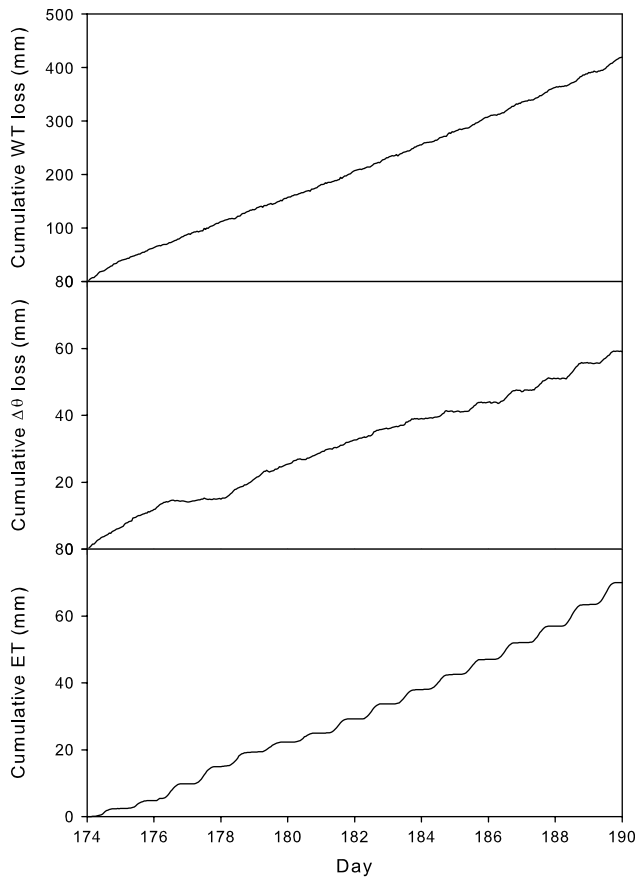


Figure 2. Cumulative water loss for ET and soil water above water table, and cumulative drop in water table depth early in the growing season during a period without rain

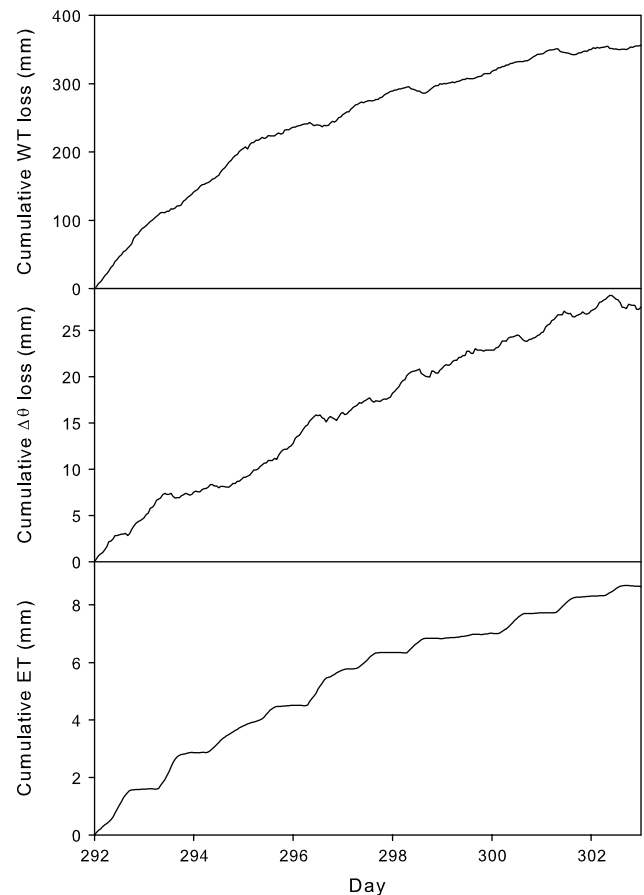


Figure 3. Cumulative water loss for ET and soil water above water table, and cumulative drop in water table depth after the growing season during a period without rain

Soil water and water table response from rainfall events

Rainfall occurring on days 173 and 190 bracketed the early crop season period, whereas rainfall on days 286 and 290 occurred before the post-growing season dry period commenced. Rain on day 173 resulted in a 2 h delay for soil wetting and water table rise (Figure 4). However, little soil wetting was evident below 0.5 m even though the water table was ~ 0.9 m. Rain on day 190 also resulted in a 2 h delay in soil wetting and water table rise, but extensive soil wetting occurred in all depth zones (Figure 5). More soil wetting appeared to occur on day 190 after the preceding 2 weeks were dry and the soil profile was dried out from crop transpiration. For the day 190 rain event, the soil water increase occurred throughout the profile within the first 3 h after the rain started (Figure 5).

Overall, during the post-growing season period, there was very little soil water increase from rain probably due to initially wet soil from previous rains and lack of plant growth to transpire and dry the soil. When wetting did occur, soil water content increased only about 1–3 mm at various depths (Figure 6). Rain on days 286–287 was slow at first, then more intense on 287 (Figure 6). When rain intensity increased, there were rapid increases in soil water and in water table rise. The water table rose from ~ 1.4 m before the rain to ~ 0.6 m 2 h after the end of the rain. Rain on day 290–291 continued for a couple days

(Figure 7). Soil water change was small because the soil was already wet. Consistent soil water increase was not apparent until 9 h after the start of rain because drainage started immediately. Soil water content averaged across depths peaked at 13 h and then declined even as the rain continued for 20 more hours. The water table rise was larger and fairly rapid starting 4–5 h after the start of rain and peaking at 20 h. The soil water drained to the saturated zone, and the saturated zone likely drained out a tile.

Specific yield estimated from rainfall events

Specific yield was estimated for four rainfall events from the ratio of RD to water level rise (Table III; Figure 8). Rainfall events were divided into smaller hourly components during days 190 and 290 to evaluate whether the Sy estimate varied across a rainfall period. Typically, RD is compared to the total amount of water level rise without considering the effects of vadose zone wetting. When this was done for two events occurring during the early growing season, Sy ranged from 0.31 to 0.46 (Table III; Figure 8). Actively transpiring plants had dried the soil so that much of the rain infiltrating into the soil wetted the vadose zone. Subtracting the water that filled the vadose zone from the RD measurement changed the Sy estimate derived from rain events during

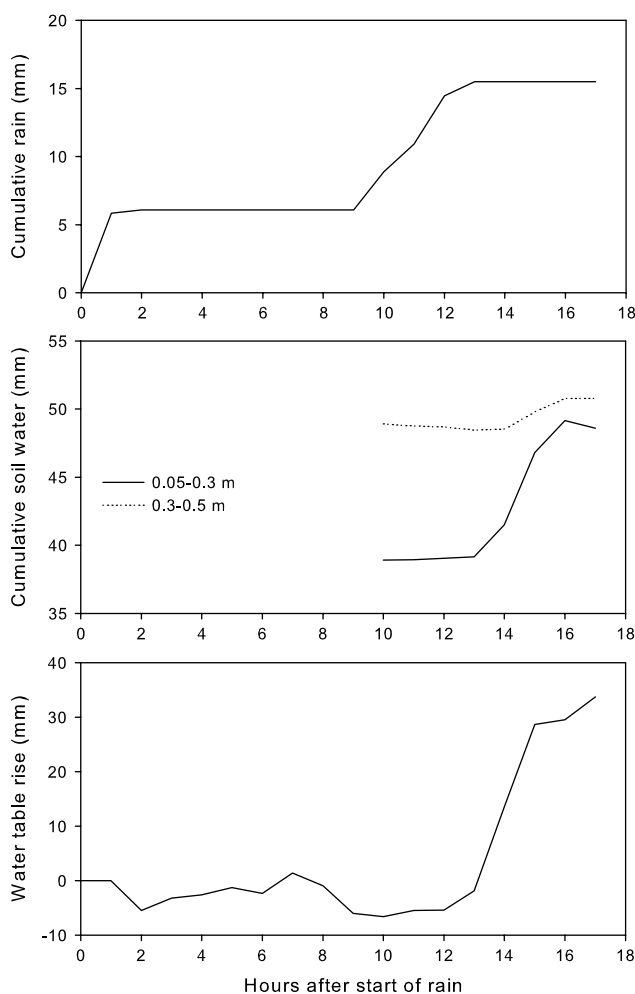


Figure 4. Day 173 rain, soil water and water table rise

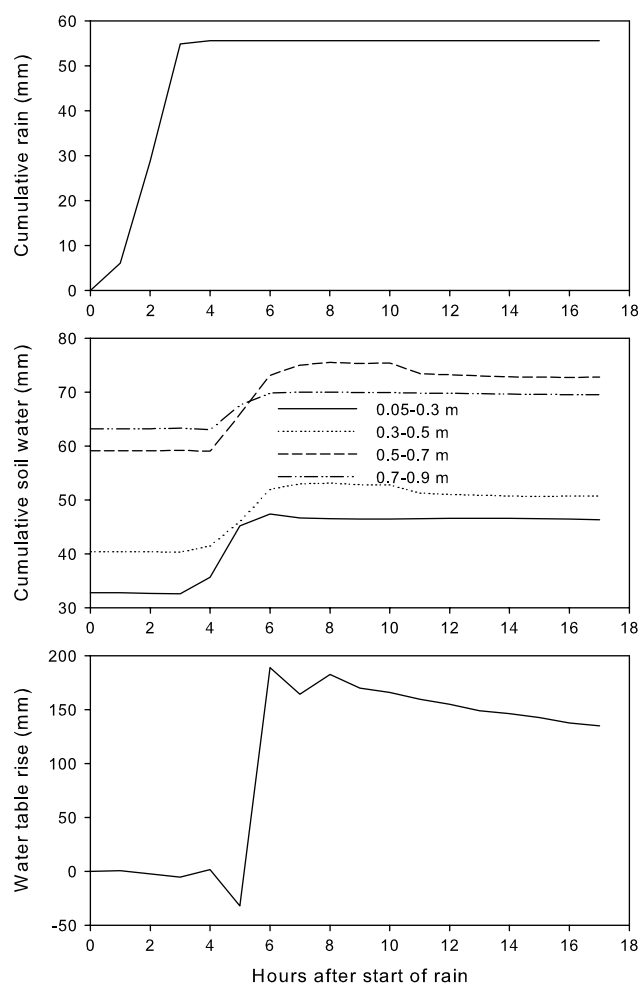


Figure 5. Day 190 rain, soil water and water table rise

the growing season (Table III; Figure 8). The mean S_y on days 173 and 190 estimated when subtracting the vadose zone wetting (0.048) was nearly an order of magnitude lower than S_y estimated using the typical method when the vadose zone wetting is ignored. During the post-growing season period, there was less difference whether or not the vadose zone wetting had been subtracted from the RD (Table III; Figure 8). The mean S_y estimated when vadose zone wetting was ignored (0.1) was slightly greater than when vadose zone wetting was subtracted (0.075), using values from days 286 and 290 (cum 31 h). When vadose zone wetting was not taken into account, the rainfall-rise method appeared to produce anomalously high S_y values, giving credence to the warnings of method applicability when soils are dry.

Comparison of methods

The mean S_y of the Webster soil was much greater when estimated during the early growing season from ET losses (0.19) compared to estimation from rainfall-water table rise during both measurement periods (0.075 when vadose zone wetting considered). When vadose zone wetting was not subtracted during the rainfall events, the estimated S_y was much greater than other estimates (>0.3). The differences in S_y values may indicate that

the estimation methods may not be measuring the same thing.

When S_y was estimated from ET, drainage from the soil profile was occurring during a sustained dry period when the water table was around 1 m in depth. These S_y values were coincidentally similar in range to the depth-compensated S_y described by Loheide *et al.* (2005). Based on the average texture of the Webster soil at the study site (44% sand, 25% clay), modelling results from Loheide *et al.* (2005) suggested that the average depth-compensated S_y for the loamy soil should be approximately 0.19, or around 0.29 for the sandy loam layer ~ 1 –1.4 m. The sand lens was within the water table for most of the two study periods examined, which probably facilitated lateral movement in the saturated zone. Unfortunately, the equation given by Loheide *et al.* (2005) did not consider how plant water uptake would extract water from soil deeper into the profile during the 2 weeks without rain because the equation only considered equilibrium soil water conditions. The dried soil just above the water table (not shown) would induce a strong gradient up from the water table, but such a gradient would not occur under equilibrium conditions.

On the other hand, the rainfall-rise method yielded a much lower S_y value (0.05). The lower S_y estimate

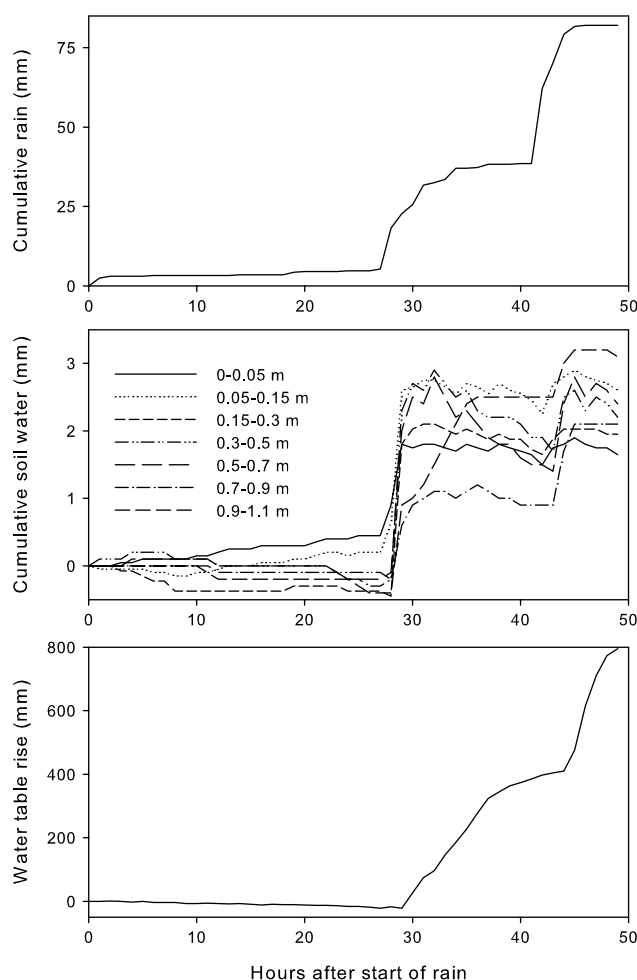


Figure 6. Day 286–287 rain, soil water and water table rise

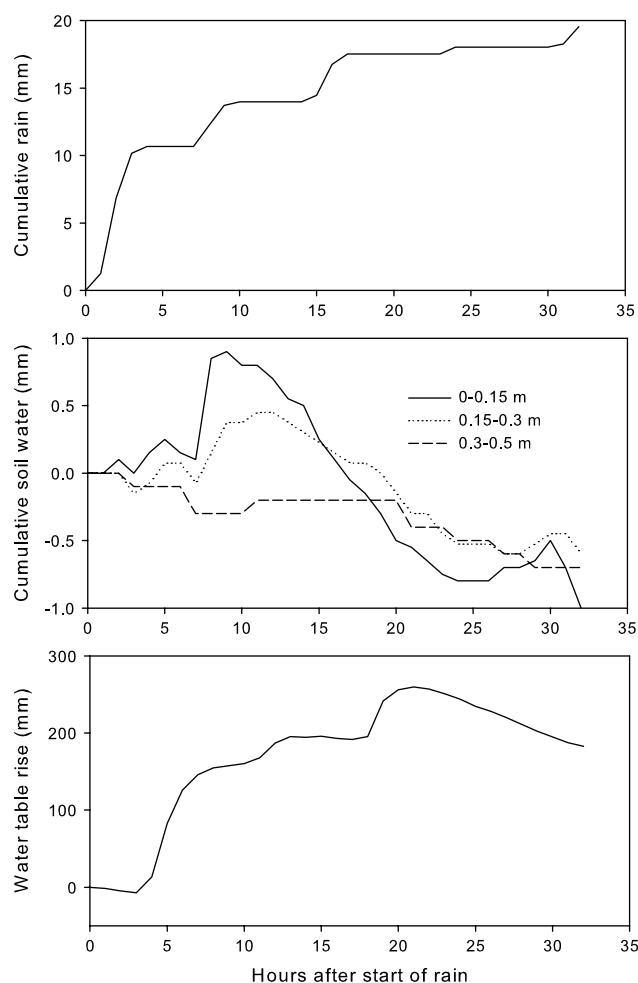


Figure 7. Day 290–291 rain, soil water and water table rise

might correspond to the 'readily available' S_y (Healy and Cook, 2002; Loheide *et al.*, 2005), which refers to the volume of water released from the vadose zone over short time periods (typically less than 12 h). The rainfall-rise method is generally not recommended when the water table depth is greater than 1 m and soil moisture is less than FC (Loheide *et al.*, 2005). In our study, the water table depth was at or near 1 m in depth when the method was applied during the growing season, and soil moisture was much less than FC as evidenced by soil wetting occurring during the rainfall events (Figures 4 and 5). However, because vadose zone wetting was measured and subtracted from our data, the rainfall-rise method was applicable for our data. The S_y estimated in our field study using the rainfall-rise method (0.05) was also similar to the modelled readily available S_y described by Loheide *et al.* (2005) for loam textured soils (~ 0.07). This would still be lower than readily available S_y for the sandy loam layer (~ 0.17), but capillary movement of water from a saturated sandy loam to an unsaturated loam should not be hindered by texture discontinuity.

The difference in S_y estimates between the ET-drainage method and rainfall-rise method might be partially attributable to the effects of air entrapment. When the water table is shallow, the capillary fringe extends

Table III. Specific yield (S_y) calculated from rain events

Day	Rain	S_y^a , ignore VZ	S_y^a , subtract VZ
	(mm)	(mm^{-1})	(mm^{-1})
173	14.4	0.461	0.065
190	55.6	0.310	0.030
190, 1 h ^b	55.6	0.296	0.023
286–287, 2 h ^b	82.0	0.100	0.083
290–291, cum 3 h ^c	10.7	0.118	0.037
290–291, cum 9 h ^c	13.6	0.064	0.048
290–291, cum 16 h ^c	17.1	0.067	0.061
290–291, cum 19 h ^c	17.2	0.064	0.053
290–291, cum 31 h ^c	18.1	0.102	0.067

^a The S_y that ignore VZ indicated the wetting of the vadose zone was not subtracted from rain. Subtract VZ indicated that wetting of the vadose zone was subtracted from rainfall amount for calculation of S_y .

^b Days 190 and 286–287 indicate calculation at maximum water table rise, which was one or 2 h after the end of the rain.

^c Days 290–291 show cumulative calculations for the long, soaking rain, for the indicated number of hours after the rain started.

into the region in which the water is flowing upward from the water table or downward to the water table (Duke, 1972). The fact that drainage S_y due to rain was less than S_y during drying may be due, in part, to trapped air during rapid rainfall (Nachabe *et al.*, 2004). For a few

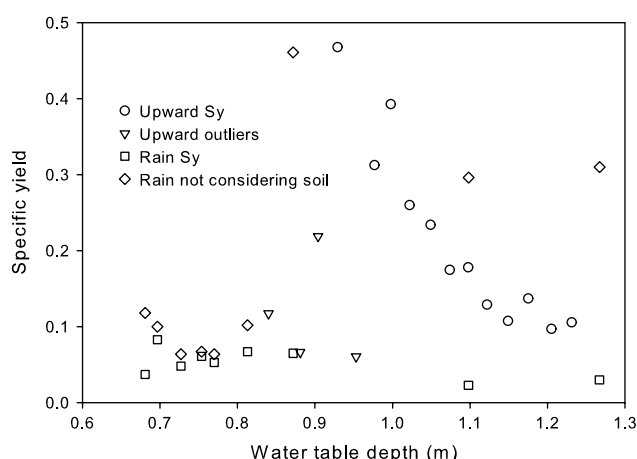


Figure 8. Specific yield (S_y) as a function of water table depth for evaporation or rain

days early after the rain on day 173, the S_y estimates were more in-line with the S_y related to rain than to the S_y related to upward water movement. Likely some drainage still occurred and there was still some trapped air. In our study, soil water contents below the water table were $0.02\text{--}0.07\text{ m}^3\text{ m}^{-3}$ less for the fall water table rise than they had been in the summer water table drop, indicating more trapped air for water table rise than for water table drop.

Effect of water table depth on S_y

The S_y patterns at our central Iowa field site showed a different response to water table depth than did other studies. The relation of S_y to water table depth varied based on the estimation method used, although it should be noted that the range in water table depth was less than 0.7 m (Figure 8). In general, with the exception of a few outliers, S_y decreased as water table depth increased when estimated using ET loss (Figure 8). S_y was greater than 0.3 when the water table was less than 1 m and decreased to 0.1 when the water table depth approached 1.3 m. A decreasing trend of S_y with increasing water table depth is contrary to trends based on an equilibrium equation (Loheide *et al.*, 2005) and field monitoring (Schilling, 2007). On the other hand, S_y did not vary by water table depth when the rainfall-rise method was used, for water table depths ranging from about 0.7 to 1.1 m (Figure 8), unless soil wetting was ignored. Ignoring soil wetting during rain showed increasing S_y as water table depth increased, but better results were obtained by considering soil wetting in the calculations.

Based on an equilibrium equation, Loheide *et al.* (2005) showed increasing S_y as depth to water table increased, especially from 0 to 1 m, and they showed increasing S_y as drainage time increased. Our data showed decreasing S_y as time increased, because our conditions were not approaching equilibrium, and most of the loss was transpiration, not drainage. Plant water uptake dried the soil at depth closer to the water table (not shown), which induced a large upward gradient. In

late June/early July, upward water loss from the water table due to ET became increasingly larger than loss due to lateral drainage. Indeed, the outliers in Figure 8 were the early days after the rain on day 173 when drainage was still predominant. The outliers more closely followed the pattern predicted by Loheide *et al.* (2005).

The models considered by Shah *et al.* (2007), Loheide *et al.* (2005) or Nachabe (2002) do not account for preferential drainage due to biopores or fractures in the soils with greater than 10% fines. These models assume that plant roots are not able to use soil water until the soil has drained to FC. Instead roots typically occupy these biopores and fractures and the roots intercept the water draining vertically through the vadose zone (Wilcox, 1962; Miller and Aarstad, 1973), and lead to more rapid cessation of vertical drainage. Lateral drainage from the saturated zone would continue for a longer time. Indeed, evaporation indirectly from a shallow saturated zone invalidates the FC concept (Cassel and Nielsen, 1986). Reicosky *et al.* (1972) have shown that plants can uptake the bulk of their water from the capillary fringe when surface soil layers are dry. The depth- and time-dependent S_y calculated by Nachabe (2002) and discussed by Loheide *et al.* (2005) is determined for a non-cropped situation, and does not consider plant water use from the lower part of the vadose zone. Shah *et al.* (2007) include plants but assume that equilibrium conditions can be re-established each day even when plants are actively taking up water.

During the post-growing season once the water table rose above 1.1 m due to rain on 286 and 287, tile drainage would commence again. Because ET loss was small, there was no competition for lateral loss to the tile. At the same time there were lateral additions from upslope perhaps even more dominant than lateral loss, which might have resulted in net soil water gain for several days after the rain ceased. Soil water on the whole was decreasing, but the calculated net effect after considering ET was soil water gain.

The simulations of Loheide *et al.* (2005) did not consider the case of lateral flow both in and out of a site, which could be responsible for the different trends in our data compared with their model. Loheide *et al.* (2005) also assumed equilibrium conditions. Our results were different from the simulations of Shah *et al.* (2007) who considered equilibrium conditions, even though they simulated shallow- and deep-rooted crops. The site considered by Schilling (2007) was quite flat, whereas the site in our experiment was $\sim 3\%$ slope. In our study, the combination of gentle slope and sandy layer at 1–1.4 m depth facilitated lateral movement into and out of the site, especially when the water table depth and capillary fringe were within the depth of the sandy layer.

CONCLUSIONS

Study results indicated that S_y was very complex. Previous research has often simplified S_y in modelling and

field studies, but our study showed that Sy varied according to ET demands, soil-water dynamics, and water table depth. We showed that Sy varied by season and method of estimation. Using the water balance method during dry periods of plant growth, Sy averaged around 0.19, which might have been inflated because of lateral flow to a nearby tile drain. Estimating Sy from rain events generally yielded a much lower value of Sy (0.02–0.08) after considering the water stored in the vadose zone. If vadose zone storage were not considered, RD estimation of Sy ranged from 0.06 to nearly 0.5, encompassing the larger Sy value estimated using the water balance. The Sy resulting from equilibrium drainage for around 1 m water table depth was not the same as the Sy resulting from water moving up from the water table into the root zone in response to soil drying due to transpiration.

Some of the key findings of this study could be generalized as follows: (i) The Sy could be estimated using a water balance method when high resolution ET and ΔS data were available; (ii) presence or absence of actively transpiring plants had a major influence on Sy; (iii) timing of groundwater recharge depended on initial soil water content and wetting of the vadose zone before the wetting front reached the water table; (iv) Sy decreased as water table depth increased; (v) Sy was lower during water table rise than during water table fall, probably due to trapped air; and (vi) integrating rainfall patterns, ET, soil water storage changes, and water table depths resulted in a useful approach for examining temporal soil water dynamics.

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